Array Programs

1. **Write a program to take an array and print the first and last element.**
2. **Write a program to take an array from the user and find:**
   a. The sum of the array.
   b. The average of the array.
3. **Write a program to print the sum of elements at even positions in an array.**
4. **Write a program to reverse an array.**
5. **Write a program to copy an array into another array.**
6. **Write a program to take an array and find the greatest (maximum) element.**
7. **Write a program to take an array and find the smallest (minimum) element.**
8. **Write a program to sort (arrange) the elements of an array in increasing order.**
   Input: 22 5 6 88 9
   Output: 5 6 9 22 88
9. **Write a program to sort (arrange) the elements of an array in decreasing order.**
   Input: 22 5 6 88 9
   Output: 88 22 9 6 5
10. **Write a program to find the second highest element in an array without sorting.**
11. **Write a program to find the second smallest element in an array without sorting.**
    Input: 22 5 6 88 9
    Output: 6
12. **Write a program to find all missing numbers from 1 to 100 in an array.**
    Example Array: {22, 17, 4, 66, 8, 2, 87}
13. **Write a program to find prime numbers from the given array:**
    Example Array: {13, 4, 56, 32, 99, 11}
14. **Write a program to print an array in the following sequence:**
    Output: 1, 2, 5, 10, 17
15. **Write a program to check whether a given element is present in an array or not.**

Multidimensional Array

1.WAP to print the sum of two matrix?
2.WAP to print the subtraction of the two matrix?
3.WAP to print Transpose matrix?

LeetCode:-

Basic Level Questions:

1. **Find the Maximum and Minimum Element in an Array**
   - Given an array, find the smallest and largest element.

Example:

Input: [3, 5, 1, 8, 2]
Output: Min = 1, Max = 8

2. **Reverse an Array**
    - Reverse a given array in-place.

Example:

Input: [1, 2, 3, 4, 5]
Output: [5, 4, 3, 2, 1]

3. **Find Second Largest Element**
    - Find the second largest number in an array without sorting.

Example:

Input: [10, 20, 4, 45, 99]
Output: 45

Intermediate Level Questions:

4. **Move All Zeros to End**
    - Move all zeros in an array to the end while maintaining the relative order of non-zero elements.

Example:

Input: [0, 1, 0, 3, 12]
Output: [1, 3, 12, 0, 0]

5. **Find Missing Number in an Array (1 to N)**
    - An array contains numbers from 1 to N with one missing number. Find it in **O(N)** time.

Example:

Input: [1, 2, 4, 5, 6]
Output: 3

6. **Find the First Non-Repeating Element**
    - Find the first element that appears only once in an array.

Example:

Input: [4, 5, 1, 2, 1, 2, 5]
Output: 4

7.Move all positive numbers to the end.
Input:[-1,12,13,0,-19,15,-10]
Output:[-1,-19,-10,0,12,15,13]

Advanced Level Questions (LeetCode Style):

8. **Product of Array Except Self (LeetCode #238)**
    - Given an array nums, return an array output such that output[i] is the product of all elements except nums[i], without using division.

Example:

Input: [1, 2, 3, 4]
Output: [24, 12, 8, 6]

9. **Find the Longest Consecutive Sequence (LeetCode #128)**
    - Find the longest consecutive sequence of numbers in a sorted array.

Example:

Input: [100, 4, 200, 1, 3, 2]
Output: 4 (Because [1, 2, 3, 4] is the longest sequence)

10. **Two Sum Problem (LeetCode #1)**
    - Find two numbers in an array that add up to a given target sum.

Example:

Input: nums = [2, 7, 11, 15], target = 9
Output: [0, 1] (Indices of 2 and 7)

11. **Find the Majority Element (LeetCode #169, Moore's Voting Algorithm)**
    - Find the element that appears more than n/2 times in an array.

Example:

Input: [3, 3, 4, 2, 4, 4, 2, 4, 4]
Output: 4

12.finds the primary and secondary diagonals of a square matrix:

Input:1 2 3
4 5 6
7 8 9

output:Primary Diagonal: 1 5 9

Secondary Diagonal: 3 5 7

13. Find Minimum in Rotated Sorted Array

Find the minimum element.

Test Cases:

Input: nums = [3,4,5,1,2]
Output: 1

Input: nums = [4,5,6,7,0,1,2]
Output: 0

Input: nums = [11,13,15,17]
Output: 11

14. Maximum Product Subarray

Find contiguous subarray with max product.

Test Cases:

Input: nums = [2,3,-2,4]
Output: 6

Input: nums = [-2,0,-1]
Output: 0

15.Best Time to Buy and Sell Stock

Find max profit from one transaction.

Test Cases:

Input: prices = [7,1,5,3,6,4]
Output: 5

Input: prices = [7,6,4,3,1]
Output: 0

15.Problem: Power Set (All Subsets)

Input: nums = [0]

Output:
```
[
 [],
 [0]
]
```

Input: nums = [1,2]

Output:
[
 [],
 [1],
 [2],
 [1,2]
]